

Detailed testing of the Black wildebeest (*Connochaetes gnou*)

Black wildebeest (*Connochaetes gnou*), or white-tailed gnu, is one of two closely related species alongside the blue wildebeest (*Connochaetes taurinus*; Grobler et al., 2018). It is endemic to southern Africa, historically occurring in South Africa, Eswatini, and Lesotho (Vrahimis et al., 2017). The species was driven extinct in Eswatini and Lesotho in the 19th century but has since been reintroduced there and introduced to Namibia, where populations are now established. It inhabits grasslands, open plains, and karoo shrublands. After a severe bottleneck of fewer than 300 individuals (Estes et al., 2004; Von Richter, 1972), the population has recovered to 16,000 individuals, including about 7,000 in Namibia (Vrahimis et al., 2016). Around 20% occur in protected areas, with most on private land. Potential threats include hybridisation with blue wildebeest (Grobler et al., 2011; Vrahimis et al., 2017). With increasing population trends, the species is currently classified as Least Concern (IUCN).

Please refer to a detailed description of evidence used for the Black wildebeest to DOI: 10.1093/biosci/biag042.

1. Fill in species and scenario information.

We tested one candidate scenario: (1) a single metapopulation.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Possible Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence

genetic

Which species are you testing? *

Connochaetes gnou

How to set up scenarios to test:

This survey allows the evaluation of **multiple subpopulation scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single metapopulation**. This one-metapopulation model must be tested first and serves as the baseline against which all alternative scenarios involving multiple subpopulations are compared.

When selecting scenarios that include **multiple subpopulations** (i.e. not the default metapopulation), **all subpopulations must be mutually distinct and supported by evidence**. Each subpopulation must have a clear evidentiary basis for inclusion in the scenario being tested. If any two subpopulations cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many Scenarios do you want to assess?

1 Scenario

1 Metapopulation

Scenario 1 name

2. Answer the questions for each scenario.

a. Genetic evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation scoring was reserved to maintain consistency across scenarios. Under this scenario, no genetic structure, fragmentation, or migration barriers are expected.

Genetic data showed no clustering (based on mitochondrial DNA, Corbet & Robinson, 1991; and microsatellite markers, Grobler et al., 2018), supporting a single metapopulation and resulting in a score of 2/2 for genetic structure.



Genetic evidence

Is there evidence of genetic structure?

Evidence through (multiple can apply): 

- Population Clustering Analyses
- Pairwise F_{ST} Values
- Private Alleles

Caution: If there is knowledge of human-mediated/controlled genetic structure or extreme bottleneck events, genetic clustering detected in a species does not automatically indicate the presences of separate subpopulations.

1 Metapopulation

☐ Yes

☒ No

☐ No data

b. Recorded biological evidence

In **Scenario (1)** (null hypothesis), populations are continuously distributed across central South Africa into Eswatini and Lesotho with ongoing gene flow (Vrahimis et al., 2016), resulting in a score of 2/2 for isolation/migration barriers.



Inferred evidence

Is there inferred evidence of likely subpopulation fragmentation?

Note: Only relevant if genetic or recorded evidence from the focal species is absent. Evidence of inferred patterns can be used. If nothing is known about the focal species, co-occurring or closely related species could give indication of subpopulation structure.

Evidence through (multiple can apply): 

- Inferred subpopulation disjunction (e.g., species distribution modeling)
- Phylogeographic and/or biogeographic evidence from co-occurring species
- Assumption of biogeographical data/refugia without direct evidence
- Inferred dispersal distance and buffer according to the CBD indicators

1 Metapopulation	2 Subpopulations	7 Subpopulations	4 Subpopulations
☐ Yes	☐ Yes	☐ Yes	☐ Yes
☐ No	☐ No	☐ No	☐ No
☒ No data	☒ No data	☒ No data	☒ No data

c. Inferred evidence

In **Scenario (1)** (null hypothesis), no evidence was found for inferred geographic fragmentation, resulting in 0/2 for this category.

Inferred evidence

Is there inferred evidence of likely subpopulation fragmentation?

Note: Only relevant if genetic or recorded evidence from the focal species is absent. Evidence of inferred patterns can be used. If nothing is known about the focal species, co-occurring or closely related species could give indication of subpopulation structure.

Evidence through (multiple can apply): ⓘ

- **inferred** subpopulation disjunction (e.g., species distribution modeling)
- Phylogeographic and/or biogeographic evidence **from co-occurring species**
- Assumption of biogeographical data/refugia **without direct evidence**
- **inferred** dispersal distance and buffer according to the CBD indicators

1 Metapopulation

☐ Yes

☐ No

☒ No data

3. Provide name and e-mail address (optional). This will allow the results to be sent to your email address.

Notes

Name & Surname

Jane Doe

Email Address

jane.doe@xxxx

Data protection disclaimer

By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

4. Click the “**Submit Data**” button to receive the results (mandatory to get results).

5. Interpreting the results.

For the three different evidence categories within Phase 1, the scenario of one metapopulation scored 4/4 and exceeded the minimum threshold of 2 points is needed, so the null hypothesis can be accepted and the Black wildebeest can be considered to represent one metapopulation and therefore also one Evolutionarily Significant Unit (ESU).

Phase 1: Identify Subpopulations

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Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Thank you , for participating in the **CGSG Subpopulation and ESU** survey.

The species you tested is the **Connochaetes gnou**

Your Survey Results:

1. Is there evidence of population structure? (2 points)

1 Metapopulation No			
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2. Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)? (2 points)

1 Metapopulation No			
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3. Is there inferred evidence of likely subpopulation fragmentation? (1 points)

1 Metapopulation No data			
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Total score for the scenario: **1 Metapopulation** is 4
Total score for the scenario: is 0
Total score for the scenario: is 0
Total score for the scenario: is 0

References

- Corbet, S. W., & Robinson, T. J. (1991). Genetic divergence in South African Wildebeest: Comparative cytogenetics and analysis of mitochondrial DNA. *The Journal of Heredity*, 82(6), 447–452. <https://doi.org/10.1093/oxfordjournals.jhered.a111126>
- Estes, R. D., Wilson, E. O., & Otte, D. (2004). *The Behavior Guide to African Mammals: Including Hoofed Mammals, Carnivores, Primates*. Berkeley: University of California Press. <https://www.google.com/search?client=firefox-b-d&q=Estes%2C+R.+D.+%282004%29.+The+Behavior+Guide+to+African+Mammals%3A+Including+Hoofed+Mammals%2C+Carnivores%2C+Primates.+Berkeley%3A+University+of+California+Press.+p.+133.+ISBN+052-0080-858>.
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